

The environmental and public health risks associated with arsenical use in animal feeds.



Arsenic exposures contribute significantly to the burden of preventable disease worldwide, specifically related to increased risks of cancer, diabetes, and cardiovascular disease. Most exposures are associated with natural contamination of groundwater, which is difficult to mitigate when these sources are used for drinking water. An anthropogenic source of arsenic exposure stems from the widespread use of arsenical drugs in food-animal production in the United States and China, among many countries. This use results in residual contamination of food products from animals raised with the drugs, as well as environmental contamination associated with disposal of wastes from these animals. Land disposal of these wastes can contaminate surface and ground water, and the conversion of animal wastes into fertilizer pellets for home use as well as the introduction of animal waste incinerators may increase opportunities for exposure. As an intentional additive to animal feed, use of arsenical drugs is a preventable source of human exposure. The domestic practice of using these drugs in poultry production has been the subject of media attention and limited research, though the use of these drugs in domestic swine production and in the rapidly growing foreign animal production industry remains largely uncharacterized. This continued expansion of arsenical drug use may likely increase the burden of global human arsenic exposure and risk.